REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions	High	Pass: Credits deduct accurately and prerequisites validated; Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed	Core functional behaviour allowing student preference input while maintaining academic eligibility.
FR-002	Functional	The system shall allow students to view, modify, and re-save their bidding credit distributions prior to the bidding deadline set by the Academic Registrar	Medium	Pass: System logs updated credit balances and saves updated preferences; Fail: Bids cannot be edited before deadline or system allows edits after cutoff.	Gives students flexibility to adjust strategies based on real-time preference updates.
FR-003	Functional	The system shall detect and prevent timetable collisions when a student attempts to bid for electives with overlapping lecture or lab schedules	High	Pass: System displays a schedule overlap warning and blocks selection; Fail: System permits bidding on courses with conflicting time slots.	Prevent invalid course combinations from entering the allocation solver stage
FR-004	Functional	The system shall execute an automated allocation algorithm to resolve seat capacities based on bid weights, prerequisite status, and student seniority	High	Pass: Allocation engine generates collision-free course lists adhering to capacity limits; Fail: Unassigned students exceed limits or timetable overlaps occur.	Automates optimal and fair distribution of limited elective seats across departments.
FR-005	Functional	The system shall provide the Academic Registrar with an administrative dashboard to open/close bidding windows and manually override edge-case allocation deadlocks.	Medium	Pass: Bidding state changes instantly upon toggle and manual overrides update course rosters; Fail: Unauthorized access permitted or status changes fail to sync	Ensures administrative oversight and operational control over the university bidding cycle
NFR-001	Non-Functional (Performance and Security)	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds	High	Pass: Benchmarking tests confirm target latency (<30s for 5,000 bids) and security standards under simulated peak load; Fail: Latency exceeds 30s or solver fails.	Guarantees timely processing during university-wide allocation without system degradation.
NFR-002	Non-Functional (Security and Availability)	The system shall encrypt all bidding data in transit and at rest using AES-256 while maintaining 99.9% uptime during open bidding windows.	High	Pass: Security audit confirms encryption standards and uptime logs show no unplanned downtime; Fail: Plaintext data detected or downtime during active bidding.	Protects student record integrity and guarantees fair, uninterrupted access during deadline hours.